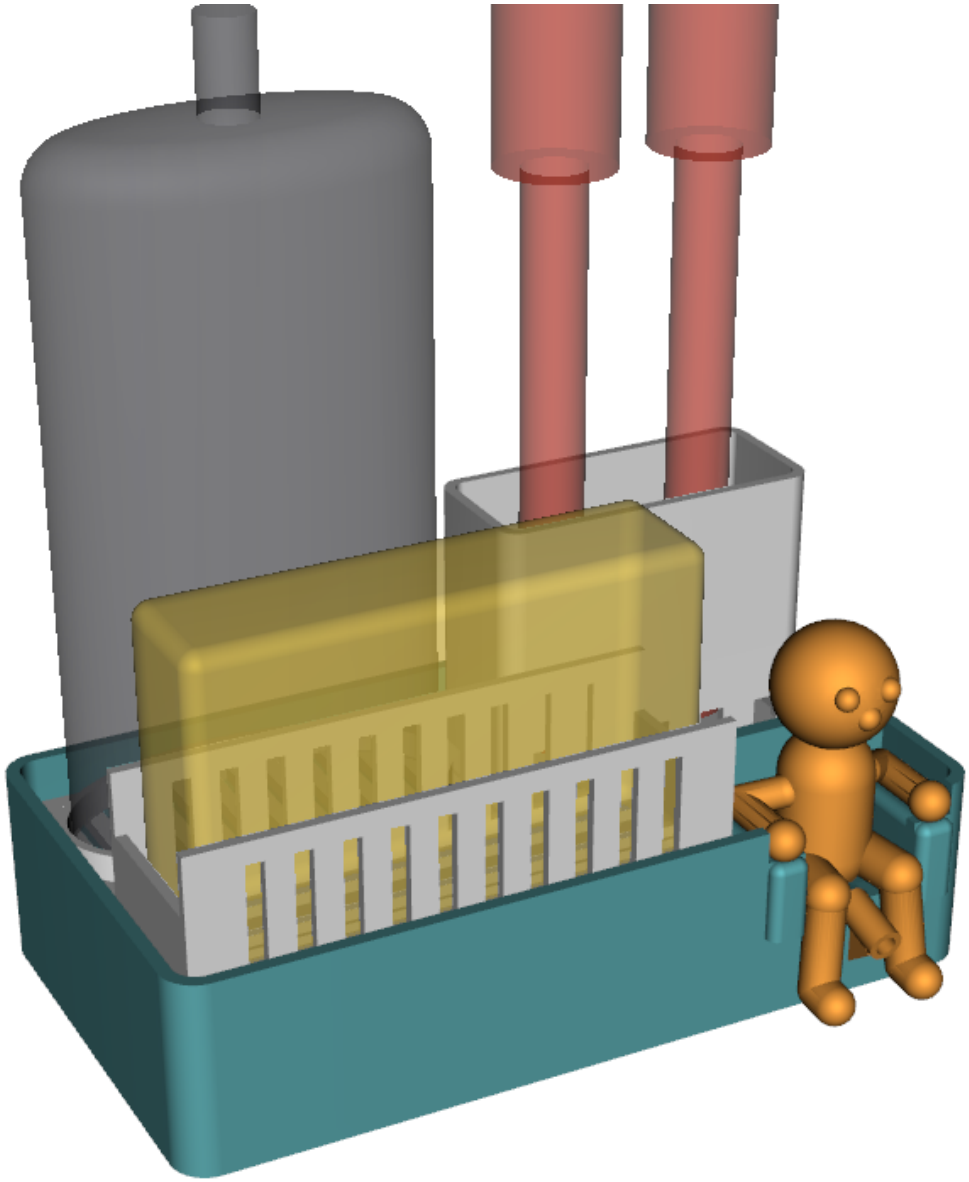
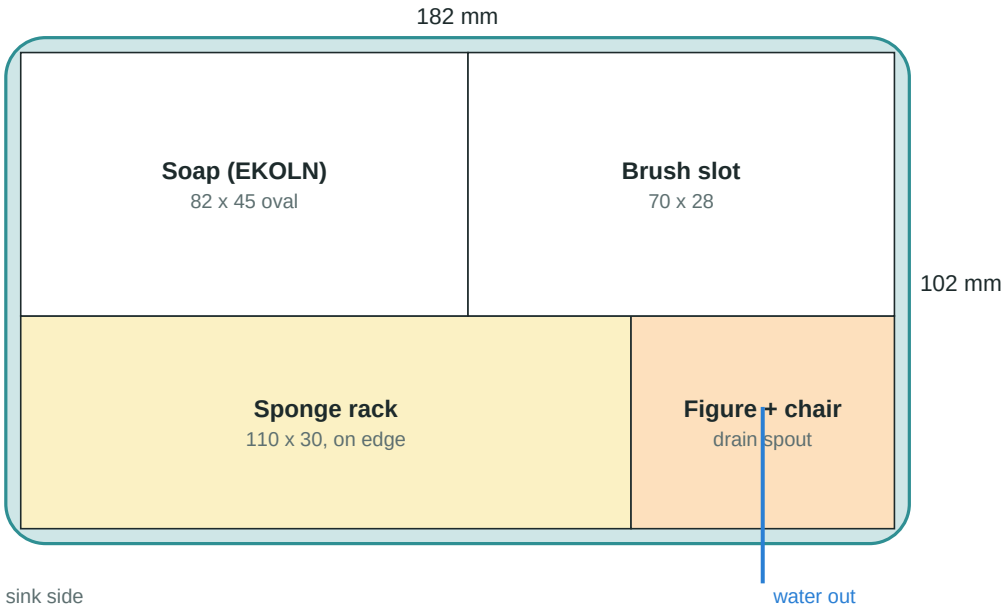
Sink Buddy

Concept v4 · 3D-printed sink organizer for a soap dispenser, brushes and a sponge, with a self-draining pan that empties through a little figure into the sink.

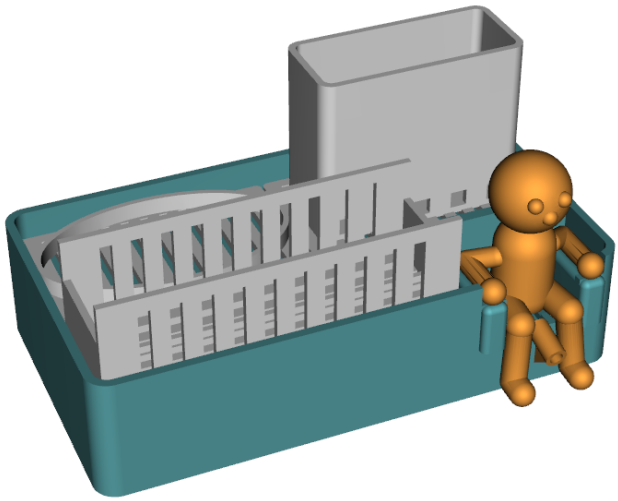


Layout (top view, to scale)

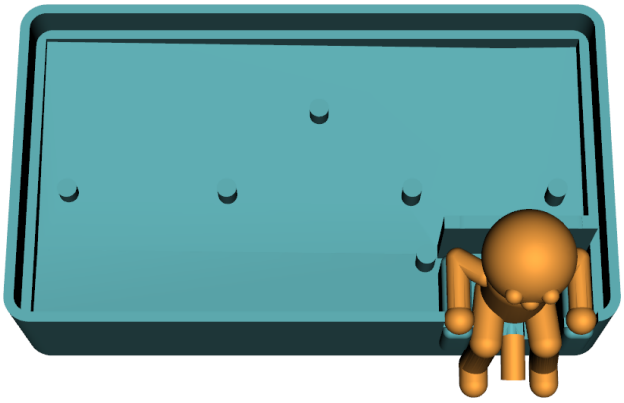

182 mm
Soap (EKOLN)
82 x 45 oval
Brush slot
70 x 28
102 mm
Sponge rack
110 x 30, on edge
Figure + chair
drain spout
sink side
water out

How it works

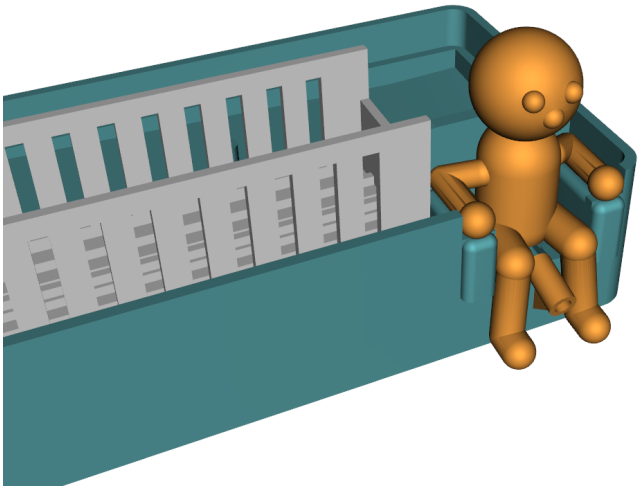
- **One shared drain pan.** All four stations sit over one pan. Its floor slopes 3° front/back and 2° sideways toward a single sump behind the figure.
- **Out through the figure.** A short tunnel runs from the sump under the figure's seat to a Ø4.5 mm spout between his knees, tilted 15° down so water drips into the sink instead of running back down the front.
- **Armchair.** The figure sits in a chair built into the front wall: armrests on both sides with his forearms on them, and a backrest bridged over a 6 mm gap so water flows underneath to the sump.
- **Lift-out inserts.** The sponge rack, soap pad and brush slot are separate drop-in parts resting on an inner shelf and posts. Lift them out to rinse the whole pan; there are no hidden channels apart from the short tunnel under the figure.
- **Only the legs stick out.** His shins, feet and the spout reach about 15 mm past the front edge; everything else is inside the footprint.



Inserts only



Drain pan with inserts removed (posts hold the inserts)



Armchair close-up

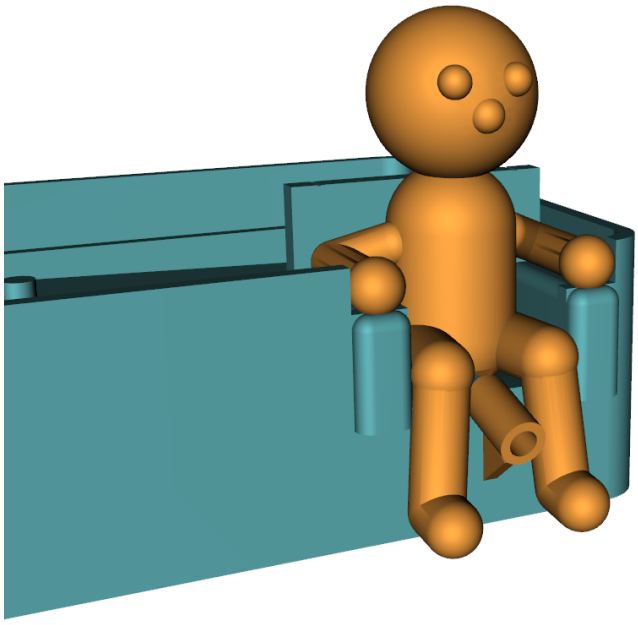


Figure and spout

Current numbers

All values are parameters in organizer.py; most are placeholders until measured.

Overall footprint	182 x 102 mm
Overall height (walls / backrest)	44 / 52 mm
Past the front edge (feet + spout)	~15 mm
Sponge, standing on edge	110 x 70 x 30 mm
Soap base (EKOLN, guessed oval)	82 x 45 mm
Brush slot, inner	70 x 28, 60 tall
Floor at drain / insert shelf	20 / 31 mm
Floor slope front-back / sideways	3 / 2 deg
Armrest height above seat	15 mm
Spout bore / tilt	4.5 mm / 15 deg
Figure height	~60 mm

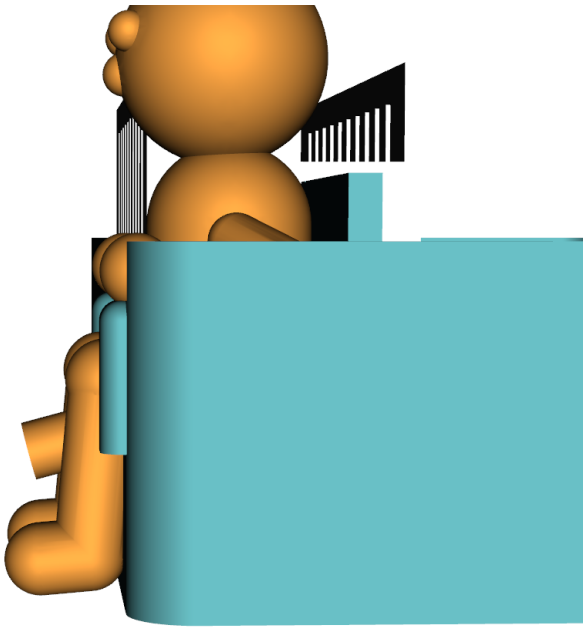
Parts to print

Base (pan, chair, posts)	1 part
Figure (separate color)	1 part
Sponge rack / soap pad / brush slot	3 inserts

Material: PETG (holds up to water and dish soap better than PLA).

Open questions

- Figure: bigger? A specific character or style?
- Figure on the right front corner, or swap with the sponge?
- Will the spout reach the sink from where it sits on the counter?
- Real measurements: sponge, EKOLN base, number and thickness of brush handles.
- Brush slot as one rectangle, or a row of round holes?



Side view: figure's legs and spout over the front edge